

Data Science

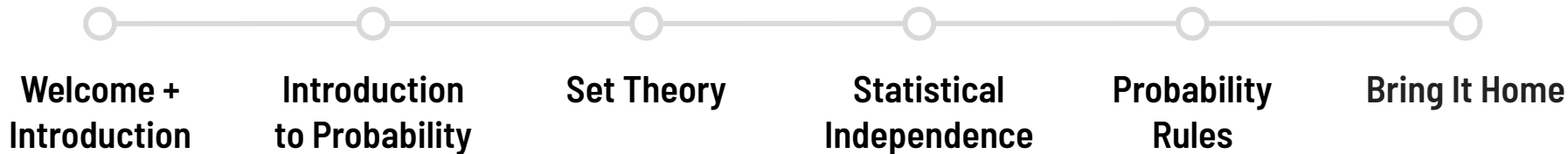
Probability

Machine Learning Fundamentals



GENERAL ASSEMBLY

Lesson roadmap



Learning objectives



1

Define experiment, outcome, event, and sample space.

2

Calculate the union and intersection of sets.

3

Apply three probability rules.

4

Solve probability problems using simulations.

Introduction to Probability

Definition

If you were to repeat an experiment infinitely many times, the **probability** of an event occurring is the proportion of times that event occurred in those experiments.

Probability is represented as a number between 0 and 1.

For example, if you flip a fair coin infinitely many times, about half of those times you will get a heads, which is a probability of 0.5.



Probability Notation

$P(A)$

Probability that...

event A occurs

Key Probability Facts

1. $P(S) = 1$: The probability that *something* happens is 1
2. $P(\emptyset) = 0$: The probability that *nothing* happens is 0
3. $0 \leq P(A) \leq 1$: The probability of any given event is between 0 and 1

Set Theory

Set Theory

Experiment

Some clearly defined “process” with an outcome. Lab coat optional.

Sample Space

The set of all possible outcomes of an experiment, usually denoted S (sometimes Ω if you're fancy).

Event

Any collection of outcomes of an experiment. A subset of the sample space.

Examples

Experiment: Flip a coin twice.

Sample Space:

Event:

Experiment: Roll one die.

Sample Space:

Event:

Examples

Experiment: Flip a coin twice.

Sample Space:

$$S = \{HH, HT, TH, TT\}$$

Event:

$$A = \{HH\}$$

$$B = \{\text{At least one H}\}$$

Experiment: Roll one die.

Sample Space:

Event:

Examples

Experiment: Flip a coin twice.

Sample Space:

$$S = \{HH, HT, TH, TT\}$$

Event:

$$A = \{HH\}$$

$$B = \{\text{At least one H}\}$$

Experiment: Roll one die.

Sample Space:

$$S = \{1, 2, 3, 4, 5, 6\}$$

Event:

$$A = \{1\}$$

$$B = \{\text{An even number}\}$$

$$C = \{\text{A prime number}\}$$

Definitions

1

Set

An unordered
collection of
distinct objects.

{ Chloe, 17, 😂 }

2

Element

An object that is
a member of a
set.

- Chloe
- 17
- 😂

Set Operations

- **Intersection:** $A \cap B$ = the set of elements in both set A **and** set B
- **Union:** $A \cup B$ = the set of elements in set A **or** in set B
- **Complement:** A^C = the set of elements **not** in A

Memory hint: AND

Examples

Experiment: Roll an 8-sided die

Sample Space: $S = \{1, 2, 3, 4, 5, 6, 7, 8\}$

$A = \{2, 4, 6, 8\}$

$B = \{2, 3, 5, 7\}$

1. **$A \cap B =$**

2. **$A \cup B =$**

3. **$A^c =$**

4. **$(A \cup B)^c =$**

Examples

Experiment: Roll an 8-sided die

Sample Space: $S = \{1, 2, 3, 4, 5, 6, 7, 8\}$

$A = \{2, 4, 6, 8\}$

$B = \{2, 3, 5, 7\}$

1. **$A \cap B = \{2\}$**
2. **$A \cup B = \{2, 3, 4, 5, 6, 7, 8\}$**
3. **$A^c = \{1, 3, 5, 7\}$**
4. **$(A \cup B)^c = \{1\}$**

Statistical Independence

Statistical Independence

The concept of independence shows up over and over again in our course.

Two events are said to be **independent** if the probability of one occurring does not affect the probability of the other occurring.

Independence

The concept of independence shows up over and over again in our course.

Two events are said to be **independent** if the probability of one occurring does not affect the probability of the other occurring.

Alternatively, A and B are **independent** if:

$$P(A \mid B) = P(A)$$



Probability that A occurs **given** that
B has already occurred

Which scenarios are independent?

	A	B
1	Flipping a heads on a coin	Rolling a 1 on a six-sided die
2	Tim hits the snooze button on his alarm clock	Someone in Paris, whom Tim does not know, hits the snooze button on his alarm clock
3	Tim has pizza for lunch	Tim has pizza for dinner
4	The amount of shark attacks on a given day is high	The amount of ice cream sales on that same day is high
5	Today's high temperature is 76°	Tomorrow's high temperature is 76°

Which scenarios are independent?

	A	B
1	Flipping a heads on a coin	Rolling a 1 on a six-sided die
2	Tim hits the snooze button on his alarm clock	Someone in Paris, whom Tim does not know, hits the snooze button on his alarm clock
3	Tim has pizza for lunch	Tim has pizza for dinner
4	The amount of shark attacks on a given day is high	The amount of ice cream sales on that same day is high
5	Today's high temperature is 76°	Tomorrow's high temperature is 76°

Which scenarios are independent?

KNOWLEDGE CHECK:

Which scenarios are independent?

	A	B	Why are these not independent?
3	Tim has pizza for lunch	Tim has pizza for dinner	If I have pizza for lunch, I will probably not have pizza for dinner.
4	The amount of shark attacks on a given day is high	The amount of ice cream sales on that same day is high	While one does <i>cause</i> the other, if the amount of shark attacks is high, it is because the weather is nice, which means ice cream sales are likely to be high
5	Today's high temperature is 76°	Tomorrow's high temperature is 76°	If it's nice today, it will more likely be nice tomorrow, too

Which scenarios are independent?

KNOWLEDGE CHECK:

Probability Rules

Rule 1: Complements

$$P(A^C) = 1 - P(A)$$

Rule 2: Intersections (*and* Probabilities)

$$P(A \cap B) = P(A)P(B \mid A)$$



Probability that A
and B occur



Probability that B
occurs **given** that A has
already occurred

Statisticians Love Urns #1

You have an urn with 12 red balls and 12 blue balls.

You pull one ball, do not put it back in, and then you pull a second ball.

What's the probability that they're both blue?

Statisticians Love Urns #1

You have an urn with 12 red balls and 12 blue balls.

You pull one ball, do not put it back in, and then you pull a second ball.

What's the probability that they're both blue?

$$P(\mathbf{B}_1 \cap \mathbf{B}_2) = P(\mathbf{B}_1) P(\mathbf{B}_2 | \mathbf{B}_1)$$

Statisticians Love Urns #1

You have an urn with 12 red balls and 12 blue balls.

You pull one ball, do not put it back in, and then you pull a second ball.

What's the probability that they're both blue?

$$P(B_1 \cap B_2) = P(B_1) P(B_2 | B_1) = (12 / 24) \times (11 / 23) = 11 / 46 = 0.239$$

Probability Calculations

Now try for yourself!

- You have an urn with 12 red balls and 12 blue balls.
- You pull one ball, do not put it back in, and then you pull a second ball.

What's the probability that they're both red?

Probability Calculations

Now try for yourself!

→ Calculate the probability:

You have an urn with 12 red balls and 12 blue balls.

You pull one ball, do not put it back in, and then you pull a second ball.

What's the probability that they're both red?

$$P(R_1 \cap R_2) = P(R_1) P(R_2 | R_1) = (12 / 24) \times (11 / 23) = 11 / 46 = 0.239$$

Rule 3: Union (*or* Probabilities)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

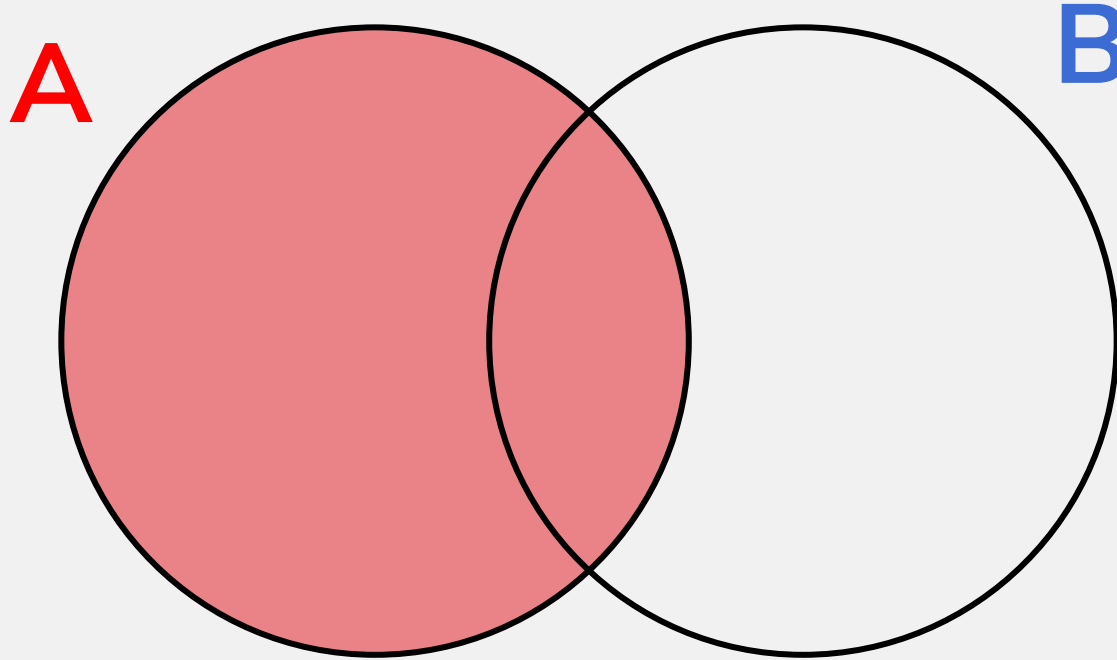


Probability that A
or B occur

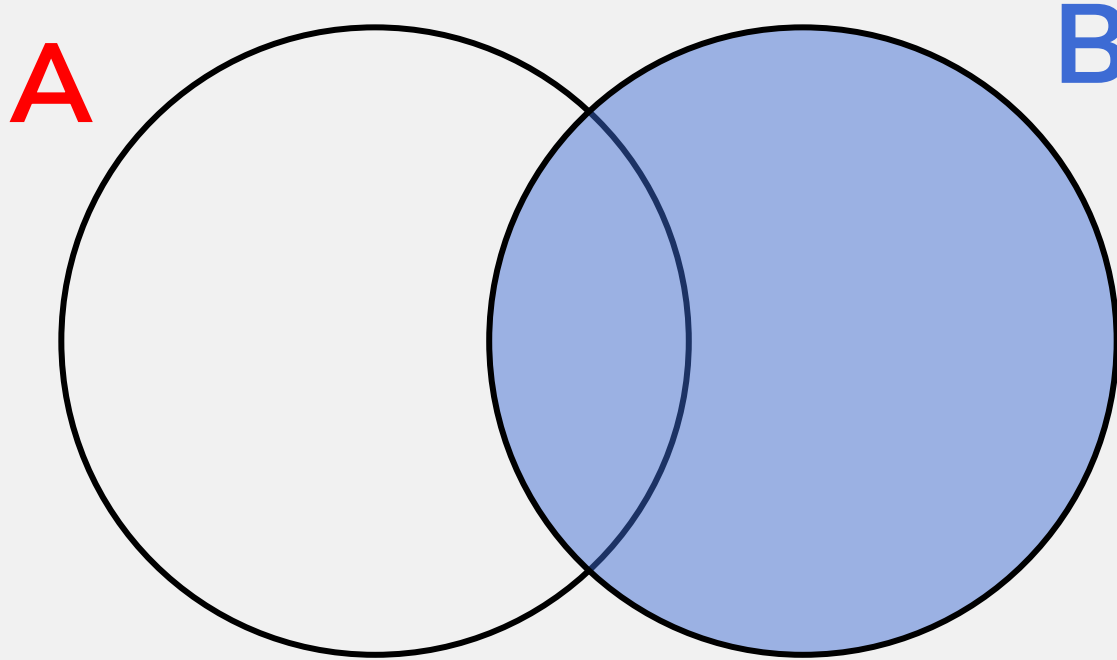


??? what???

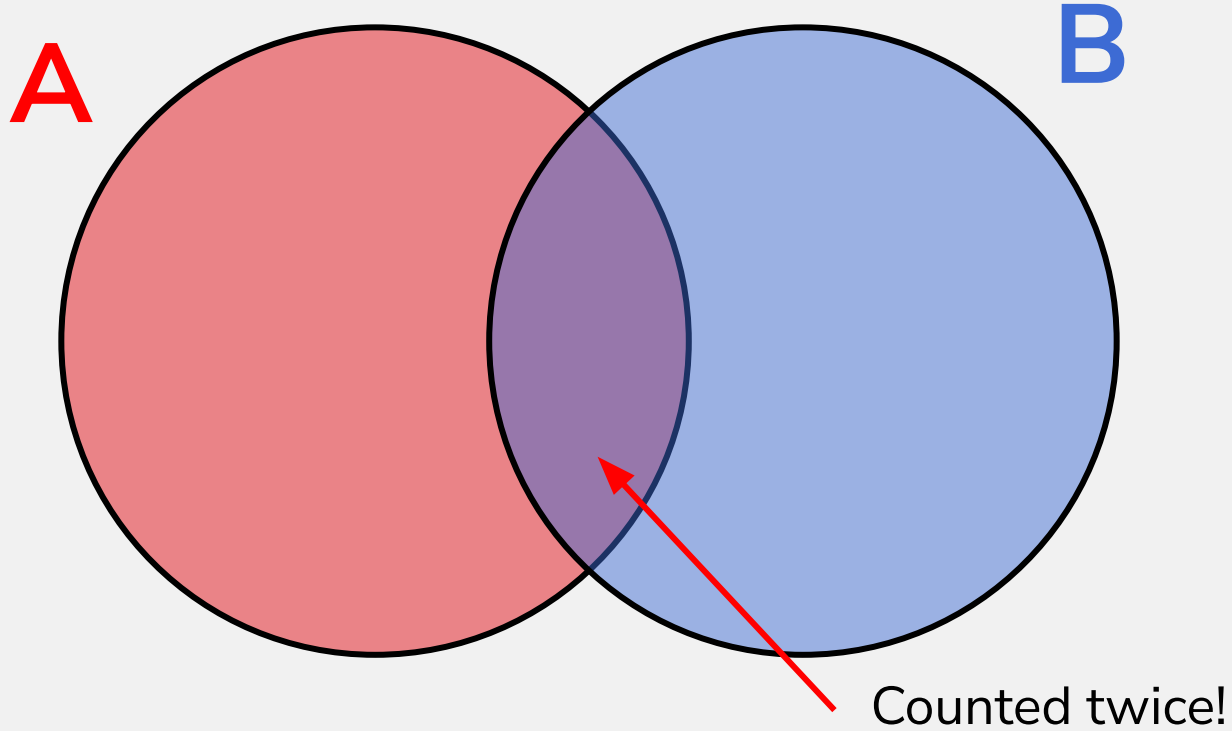
Probability Unions



Probability Unions



Probability Unions



Statisticians Love Urns #2

You have an urn with 12 red balls and 12 blue balls.

You pull one ball, do not put it back in, and then you pull a second ball.

What's the probability that they're both the same color?

Statisticians Love Urns #2

You have an urn with 12 red balls and 12 blue balls.

You pull one ball, do not put it back in, and then you pull a second ball.

What's the probability that they're both the same color?

$$P(\text{Both Blue} \cup \text{Both Red}) = P(\text{Both Blue}) + P(\text{Both Red}) - P(\text{Both Blue} \cap \text{Both Red})$$

Statisticians Love Urns #2

You have an urn with 12 red balls and 12 blue balls.

You pull one ball, do not put it back in, and then you pull a second ball.

What's the probability that they're both the same color?

$$P(\text{Both Blue} \cup \text{Both Red}) = P(\text{Both Blue}) + P(\text{Both Red}) - P(\text{Both Blue} \cap \text{Both Red})$$

These events are **disjoint!**

Probability Calculations

Now try for yourself!

→ Calculate the probability:

You have an urn with 12 red balls and 12 blue balls.

You pull one ball, do not put it back in, and then you pull a second ball.

What's the probability that they're both the same color?

$$P(\text{Both Blue} \cup \text{Both Red}) = P(\text{Both Blue}) + P(\text{Both Red}) =$$

Calculate the probability

Now try for yourself!

→ Calculate the probability:

You have an urn with 12 red balls and 12 blue balls.

You pull one ball, do not put it back in, and then you pull a second ball.

What's the probability that they're both the same color?

$$P(\text{Both Blue} \cup \text{Both Red}) = P(\text{Both Blue}) + P(\text{Both Red}) = 0.239 + 0.239 = 0.478$$

When by hand is tough...

When these problems get too hard, why do the math?

We can often find the approximate answers via **simulation**.

Why does this work?

The Law of Large Numbers (LLN)

The **Law of Large Numbers** states that if you were to repeat an experiment infinitely many times **and some other conditions which are always met in practice**, then the simulated probability of an event approaches the true probability.*

**Ok, the real LLN is more complicated and has many more implications than this, and there are many different kinds of LLNs. This is a good gist, though.*

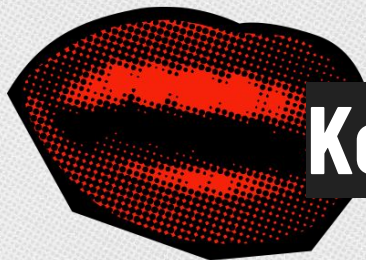
Probability

Let's see it for ourselves!

- Open **02-probability.ipynb**



Bring It Home



Key takeaways



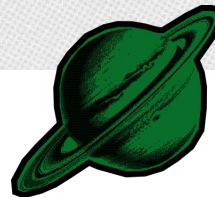
Probability is a key building block

Probability is the foundation of machine learning.



Rules of set theory guide your work

From merging sql tables to large language models rules of probability and set theory are widely used.



Stuck on a logic problem?

Come back and review probability!

DON'T FORGET:

Exit tickets



